

ROMANOVKA, Russian Federation

WMO: 306500

Lat: 53.200N

Lon: 112.783E

Elev: 3027

StdP: 13.16

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -41.0	(c) -36.6	(d) -45.4	(e) 0.4	(f) -40.6	(g) -41.0	(h) 0.6	(i) -36.3	(j) 18.3	(k) -3.6	(l) 15.2	(m) -2.3	(n) 0.1	(o) 230	(p) 0.664

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 23.5	(c) 82.2	(d) 61.4	(e) 78.6	(f) 60.3	(g) 75.2	(h) 59.0	(i) 64.7	(j) 74.9	(k) 63.0	(l) 72.8	(m) 61.5	(n) 71.0	(o) 4.5	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.8	92.5	67.9	59.9	86.3	65.8	58.0	80.6	64.5	31.6	74.5	30.2	73.2	29.0	71.1	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 14.5	(b) 12.2	(c) 10.5	(e) -47.0	(f) 87.5	(g) 6.1	(h) 3.8	(i) -51.3	(j) 90.3	(k) -54.9	(l) 92.5	(m) -58.3	(n) 94.7	(o) -62.7	(p) 97.5		
(a) 14.5	(b) 12.2	(c) 10.5	(e) -46.4	(f) 68.0	(g) 5.4	(h) 2.0	(i) -50.3	(j) 69.4	(k) -53.4	(l) 70.6	(m) -56.4	(n) 71.8	(o) -60.4	(p) 73.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.5	-14.2	-5.3	10.9	30.0	43.9	57.0	62.2	56.9	43.7	27.4	3.5	-11.7		
	DBStd	28.41	10.68	10.59	11.84	9.26	6.97	6.49	5.15	5.50	7.36	9.23	13.10	11.88		
	HDD50	9815	1991	1548	1212	601	214	15	0	11	216	701	1394	1912		
	HDD65	14451	2456	1968	1677	1050	655	249	117	254	638	1166	1844	2377		
	CDD50	881	0	0	0	1	24	225	378	225	28	0	0	0		
	CDD65	41	0	0	0	0	0	8	30	3	0	0	0	0		
	CDH74	965	0	0	0	0	17	287	521	129	11	0	0	0		
	CDH80	202	0	0	0	0	2	58	122	20	0	0	0	0		
	WSAvg	3.4	2.8	3.2	3.6	4.6	4.8	3.4	2.6	2.7	3.5	3.6	3.2	2.7		
	PrecAvg	14.3	0.2	0.1	0.2	0.8	1.4	2.0	3.7	3.4	1.7	0.4	0.3	0.2		
(15) Precipitation	PrecMax	21.2	0.4	0.3	0.6	1.7	4.0	4.7	7.2	5.4	3.5	1.2	0.8	0.5		
	PrecMin	9.3	0.0	0.0	0.1	0.2	0.4	0.3	0.7	1.4	0.4	0.0	0.1	0.0		
	PrecStd	2.7	0.1	0.1	0.1	0.4	0.7	1.1	1.7	1.1	0.8	0.3	0.2	0.1		
	PrecStd	2.7	0.1	0.1	0.1	0.4	0.7	1.1	1.7	1.1	0.8	0.3	0.2	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	28.7	47.4	64.3	76.9	85.5	87.7	82.6	75.0	58.7	39.2	22.2		
		MCWB	14.9	22.9	35.7	45.2	55.0	61.1	63.8	61.2	55.0	44.6	31.3	19.3		
	2%	DB	10.7	22.7	41.1	57.1	71.3	80.9	83.5	77.4	69.4	52.1	32.9	15.9		
		MCWB	8.6	18.2	31.9	41.8	51.3	59.8	62.7	59.9	53.0	40.3	28.0	13.7		
	5%	DB	6.4	17.8	36.0	51.8	66.3	77.0	79.9	73.5	63.6	46.7	28.3	11.1		
		MCWB	4.9	14.5	28.2	39.0	48.0	58.3	62.2	59.0	50.0	37.1	24.5	9.6		
	10%	DB	2.5	13.5	31.1	46.9	61.7	73.3	76.2	69.9	58.8	42.1	23.6	6.3		
		MCWB	1.3	10.9	25.0	36.0	46.1	56.8	61.0	58.5	48.0	34.3	20.7	5.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	23.2	36.1	45.9	55.6	65.7	67.7	65.6	57.1	44.9	32.1	19.4		
		MCDB	17.2	28.7	46.8	62.5	75.7	76.5	77.8	72.7	68.4	58.4	38.3	22.2		
	2%	WB	8.8	18.3	31.9	42.4	52.0	62.6	65.6	62.9	54.9	40.8	28.3	13.7		
		MCDB	10.5	22.4	40.6	56.0	68.5	73.0	76.3	70.3	65.6	50.6	33.0	15.6		
	5%	WB	5.1	14.6	28.5	39.5	49.2	60.5	64.0	61.4	52.2	37.7	24.6	9.6		
		MCDB	6.5	17.7	35.6	51.2	64.3	71.7	74.8	69.5	61.4	45.8	28.2	11.0		
	10%	WB	1.4	11.0	25.2	36.3	46.8	58.6	62.6	59.7	49.4	34.8	20.7	5.2		
		MCDB	2.5	13.5	31.0	46.0	59.9	70.1	73.0	67.9	57.1	41.5	23.5	6.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.7	29.3	30.4	24.0	25.9	25.9	23.5	22.6	24.6	22.8	23.1	20.5		
		MCDBR	21.8	29.7	31.5	32.1	36.4	34.7	31.5	30.7	34.3	29.7	23.5	18.8		
		MCWBR	20.2	25.7	23.2	19.5	19.1	15.2	13.2	14.9	18.1	18.7	18.7	17.2		
		MCDBR	21.6	29.4	30.8	30.7	33.8	27.0	23.3	23.2	29.2	28.3	23.1	18.8		
	5% WB	MCWBR	20.0	25.6	22.8	19.1	18.2	12.7	11.0	12.6	15.9	18.2	18.5	17.3		
(40) Clear-Sky Solar Irradiance	taub		0.237	0.264	0.296	0.376	0.480	0.451	0.395	0.367	0.322	0.282	0.258	0.237		
	taud		2.220	2.201	2.203	2.143	1.900	2.098	2.335	2.394	2.469	2.482	2.307	2.185		
	Ebn at Noon		241	268	282	270	245	254	269	271	273	263	231	214		
	Edh at Noon		19	28	35	43	57	48	37	33	27	21	18	17		
(44) All-Sky Solar Radiation	RadAvg		402	759	1285	1633	1774	1901	1759	1487	1149	758	424	299		
	RadStd		13	22	38	99	95	127	161	95	80	50	26	15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-2.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page